

GROUP NUMBER - 05

# 3D RECONSTRUCTION FROM 2D PHOTOS AND DATA

## Utilizing Neural Radiance Fields (NeRF) and Gaussian Splatting

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### DATE

05 January 2026

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## 04: Objective

Framework development for hybrid 3D generation and performance optimization.



## 05: Conclusion

Future outlook and summary of finding in neural radiance and splatting fields.

# LITERATURE REVIEW (1-3 / 20)

| SI | Title of Paper                                                                        | Publisher                                      | Methodology                                                                                                                                                                                                                                                                                   | Year | Advantages                                                                                                                                                                                  | Disadvantages                                                                                                                                                                                                                                                                                         |
|----|---------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | gsplat: An Open-Source Library for Gaussian Splatting                                 | Journal of Machine Learning Research 26 (2025) | <b>Gaussian Splatting (PyTorch + CUDA):</b> Open-source library featuring <b>CUDA-optimized rasterization, densification, differentiable pose optimization, depth rendering, N-D splatting, and Mip-Splatting anti-aliasing.</b>                                                              | 2025 | Achieves up to 10% less training time and 4x less memory compared to the original Kerbl et al. (2023) implementation.                                                                       | High GPU memory usage, CUDA/PyTorch dependency issues, fill-rate-heavy rendering, sensitive hyperparameters, limited scalability for very large scenes                                                                                                                                                |
| 2  | A Survey on 3D Gaussian Splatting Applications: Segmentation, Editing, and Generation | arXiv:2508.09977v1 [cs.CV]                     | <b>3D Gaussian Splatting Survey:</b> Comprehensive review of 3DGS methods and applications ( <b>Segmentation, Editing, Generation, Avatars, SLAM</b> ), analyzing connections with <b>NeRF</b> and <b>2D foundation models.</b>                                                               | 2025 | Real-time photorealistic rendering with explicit, interpretable representation enabling efficient optimization and intuitive high-level manipulation                                        | Lack of faithful 3D-aware evaluation metrics, scarcity of large-scale annotated 3D data, and inefficient per-scene optimization                                                                                                                                                                       |
| 3  | A review on 3D Gaussian splatting for sparse view reconstruction                      | Artificial Intelligence Review                 | <b>Sparse Inputs</b> (Single-view, Few-shot), <b>Multi-dim. Feature Extract</b> (Point Cloud (PC), Depth Map, Semantic, Spatial), <b>3D GS Generation</b> (End-to-end, Diff. Models, Transformer), and <b>Anomaly Handling</b> (Artifact/Scale Ambiguity mitigation). Compares to <b>NeRF</b> | 2025 | Superior <b>reconstruction quality</b> and high <b>efficiency/real-time rendering</b> compared to NeRF-based algorithms. Explicit representation reduces overhead in low-density scenarios. | Prone to <b>overfitting</b> in sparse views. Common issues include <b>artifacts</b> (floaters) and <b>scale ambiguity</b> . Struggles with <b>complex textures</b> and <b>special scenes</b> (e.g., specular reflections). Limitations of discrete <b>Gaussian representations</b> in dynamic scenes. |

# LITERATURE REVIEW (4-6 / 20)

| SI | Title of Paper                                                                                     | Publisher                                                              | Methodology                                                                                                                                                                                                                     | Year | Advantages                                                                                                                                                                                                                        | Disadvantages                                                                                                                                                                                          |
|----|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | COB-GS: Clear Object Boundaries in 3DGS Segmentation Based on Boundary-Adaptive Gaussian Splitting | CVPR (Open Access version, provided by the Computer Vision Foundation) | <b>Semantic-Aware 3DGS</b> – Joint optimization of semantics and visuals using Boundary-Adaptive Gaussian Splitting (BAGS), Boundary-Guided Texture Restoration (BGTR), and a two-stage mask generation pipeline based on SAM2. | 2025 | Substantially improves <b>segmentation accuracy</b> and <b>robustness</b> against inaccurate masks. Yields <b>clear boundaries</b> while <b>preserving high visual quality</b> .                                                  | Current work is subject to <b>floating artifacts</b> which are magnified after segmentation (future work focus).                                                                                       |
| 5  | EVSplitting: An Efficient and Visually Consistent Splitting Algorithm for 3D Gaussian Splatting    | SA Conference Papers                                                   | <b>EVSplitting: Algorithm for 3DGS splitting that solves two minimization problems: ensuring Visual Consistency and reducing Gaussian Overflow, used for explicit operations like engraving and point cloud uniformization.</b> | 2025 | <b>Efficient</b> and <b>visually consistent</b> split. Produces <b>clearer boundaries</b> and consistent visual appearance. <b>Faster</b> than Gaussian Mixture Models (GMMs). Leads to <b>denser/more uniform Point Clouds</b> . | <b>Straightforward splitting</b> causes blurs, bulges, and holes. <b>Floating point precision errors</b> can accumulate over continuous splitting/editing operations.                                  |
| 6  | NeRF: Neural Radiance Field in 3D Vision: A Comprehensive Review                                   | ECCV                                                                   | Implicit, neural network-based scene representation using differentiable volume rendering with MLPs.                                                                                                                            | 2025 | Revolutionized Computer Vision; self-supervised (only requires multi-view images and poses); produces state-of-the-art visual quality; low storage and memory requirements.                                                       | Fully implicit methods take two to three orders of magnitude longer to train; fully implicit methods render slower by orders of magnitude; extracting an explicit 3D representation is more difficult. |

# LITERATURE REVIEW (7-9 / 20)

| SI | Title of Paper                                                                                                   | Publisher                                    | Methodology                                                                                                                                                         | Year | Advantages                                                                                                                                                                                                                                                                                                | Disadvantages                                                                                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | Neural Pruning for 3D Scene Reconstruction: Efficient NeRF Acceleration                                          | arXiv                                        | Studies neural pruning strategies (uniform sampling, importance-based, and coreset-based techniques) to reduce the model size and speed up training of NeRF models. | 2025 | Coreset-driven pruning can achieve a <b>50% reduction in model size</b> and a <b>35% speedup in training</b> .                                                                                                                                                                                            | NeRF models demand lengthy training times, often spanning days. Pruning via coreset methods results in only a slight decrease in accuracy.                              |
| 8  | DT-NeRF: A Diffusion and Transformer-Based Optimization Approach for Neural Radiance Fields in 3D Reconstruction | ICCK Transactions on Intelligent Systematics | Proposes a Diffusion Model-Optimized Neural Radiance Field (DT-NeRF) method by combining diffusion models with Transformers.                                        | 2025 | Significantly outperforms traditional NeRF and other state-of-the-art methods on datasets like Matterport3D and ShapeNet in metrics such as PSNR, SSIM, Chamfer Distance, and Fidelity. Effective at restoring details under sparse viewpoints and maintaining high accuracy in complex geometric scenes. | Increased computational cost and time due to joint optimization and extra modules, which challenges scalability to large-scale, dynamic scenes and real-time rendering. |
| 9  | High-Precision Visual SLAM for Dynamic Scenes Using Semantic–Geometric Feature Filtering and NeRF Maps           | Electronics (MDPI)                           | SLAM integrates: Image Enhancement, Quantized Feature Detection, Semantic-Geometric Dynamic Filtering, and NeRF-based Static Reconstruction.                        | 2025 | Achieves <b>93.4% average RMSE reduction</b> vs. ORB-SLAM3. High accuracy, robustness, and adaptability.                                                                                                                                                                                                  | Primarily evaluated on indoor data; limited generalization to large-scale outdoor or rapid motion scenarios.                                                            |

# LITERATURE REVIEW (10-12 / 20)

| SI | Title of Paper                                                                                     | Publisher                                                   | Methodology                                                                                                                                                                                          | Year | Advantages                                                                                                                                                                                                                                   | Disadvantages                                                                                                                                                                       |
|----|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | AE-NeRF: Augmenting Event-Based Neural Radiance Fields for Non-ideal Conditions and Larger Scenes  | The Thirty-Ninth AAAI Conference on Artificial Intelligence | <b>AE-NeRF Framework</b> – Joint pose–NeRF optimization with pose correction, hierarchical event distillation for large scenes, normalized multi-loss training, and learning-based color correction. | 2025 | State-of-the-art event-based 3D reconstruction, robust to pose errors and sparse events, scales to large scenes, and delivers sharper renderings with cleaner geometry                                                                       | The notable improvement demonstrated by the method is <b>less pronounced in object-level scenes</b> compared to its performance in large-scale scenes.                              |
| 11 | Flow-NeRF: Joint Learning of Geometry, Poses, and Dense Flow within Unified Neural Representations | CVPR                                                        | Unified optimization of geometry, poses, and dense optical flow using pose-conditioned bijective flow mapping, flow-to-geometry feature message passing, and shared point sampling                   | 2025 | Outperforms prior methods in novel-view synthesis and depth estimation, enables pose-free novel-view image/depth/flow inference, produces accurate long-range flow, and significantly improves geometry via flow-based feature distillation. | Assumes static scenes (camera-induced flow only), depends on RAFT-based pseudo-flow supervision, and achieves pose estimation comparable to (but not surpassing) Nope-NeRF overall. |
| 12 | A Survey of 3D Reconstruction: The Evolution from Multi-View Geometry to NeRF and 3DGS             | Sensors(MDPI)                                               | <b>3D Reconstruction Survey</b> – Review of SfM/MVS, NeRF, and 3DGS with comparative analysis of quality, efficiency, sparse-view support, and dynamic scenes.                                       | 2025 | Comprehensive researcher-oriented survey tracing evolution from SfM/MVS to NeRF and 3DGS, providing multi-dimensional comparison of strengths and limitations, and outlining key challenges and future research directions.                  | High memory/storage cost for large-scale scenes, dependence on accurate SfM camera poses, and limited stability in dynamic scenes and cross-scene generalization.                   |

# LITERATURE REVIEW (12-15 / 20)

| SI | Title of Paper                                                                                         | Publisher                       | Methodology                                                                                                                                                                                                                        | Year | Advantages                                                                                                                                                                                                   | Disadvantages                                                                                                                                                                                                                                                    |
|----|--------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | Specularity in NeRFs: A Comparative Study of Ref-NeRF and NRFF                                         | Image Processing On Line (IPOL) | Evaluation of Ref-NeRF and NRFF for specular surface rendering, comparing quality and convergence on simple and complex reflective datasets.                                                                                       | 2025 | Both outperform NeRF/Mip-NeRF on specular surfaces; Ref-NeRF excels in accuracy and handling geometry errors, while NRFF converges faster                                                                    | Struggle with inter-reflections, rely on altered normals for anisotropic specularities, NRFF poorly models diffuse/specular colors and allows negative radiance, Ref-NeRF can misassign specularity, and both assume single-bounce lighting without ray tracing. |
| 14 | SC-NeRF: NeRF-based Point Cloud Reconstruction using a Stationary Camera for Agricultural Applications | CVPR                            | Single fixed-camera capture of a rotating object, COLMAP-based pose estimation with pose transformation to simulate camera motion, ROI-based scene filtering, followed by standard NeRF training.                                  | 2025 | Produces near-perfect PCDs, avoids moving-camera setups, maintains efficient training, and supports high-throughput phenotyping with delicate sensors.                                                       | Pose estimation is the main computational bottleneck, preprocessing is slower than GT moving-camera setups, pose optimization needs improvement, and broader validation across plant species and phenotyping scenarios is required.                              |
| 15 | GO-N3RDnet: Geometry Optimized NeRF-enhanced 3D Object Detector                                        | CVPR                            | End-to-end multi-view 3D object detection with PEOM for positional voxel embedding, DIS for foreground-focused sampling using voxel/NeRF densities, and OOM for cross-view opacity consistency via ray-length-weighted refinement. | 2025 | Sets new SOTA for NeRF-based multi-view 3D detection on ScanNet and ARKitScenes, +10.2% mAP@0.25 over ImVoxelNet, matches CN-RMA with ~14× fewer epochs and smaller grids, and needs no GT TSDF supervision. | Complex architecture with multiple custom modules, performance only on par with CN-RMA despite added complexity, and dependence on a pre-trained 2D backbone (e.g., ResNet-50/101                                                                                |



# LITERATURE REVIEW (16-18 / 20)

| SI | Title of Paper                                                                                                     | Publisher                                            | Methodology                                                                                                                                                                                       | Year | Advantages                                                                                                                                                                                                               | Disadvantages                                                                                                                                                                                                                              |
|----|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 | Comparative Analysis of 3D Reconstruction Results Using MVS and NeRF:Insights from PCA                             | IOP Publishing Ltd                                   | Comparative 3D reconstruction using MVS and NeRF on identical image sets, with PCA-based quantitative and qualitative evaluation of accuracy, geometric consistency, and data variance.           | 2025 | High-detail reconstruction, robust handling of complex geometry, objective PCA-based evaluation, and suitability for environmental and 3D analysis                                                                       | High compute and training cost, limited scalability to large scenes, dataset-dependent performance, and reliance mainly on PCA for evaluation.                                                                                             |
| 17 | Overview of NeRF Technology and Applications                                                                       | Journal of Computer-Aided Design & Computer Graphics | Comprehensive review of NeRF methods, categorizing approaches by pose estimation, view aggregation, and rendering optimization, with analysis of similarities, differences, and application scena | 2025 | Highlights NeRF's 2D-to-3D strengths, analyzes optimization methods (pose and non-rigid motion), outlines future research directions, and serves as a clear introductory guide.                                          | Slow rendering due to heavy computation, limited quality in single/sparse-view setups, poor cross-scene generalization, inefficient conversion to standard 3D formats (e.g., BIM/3DMAX), and lack of accessible internal scene information |
| 18 | Toward Automatic 3D Model Reconstruction of Building Curtain Walls from UAV Images Based on NeRF and Deep Learning | Remote Sensing (MDPI)                                | UAV image–based BIM reconstruction using NeRF for point cloud generation, PointNet++ for semantic segmentation, geometric parameter extraction, and automated BIM modeling via Revit + Dynamo.    | 2025 | Achieves <0.1 m BIM dimensional error, outperforms photogrammetry for curtain walls, improves semantic segmentation over traditional methods, and enables automatic semantic BIM reconstruction missing in standard NeRF | Limited generalization due to same-building train/test data, lack of comparison with other segmentation networks (e.g., PointCNN), and need for higher-quality cameras for higher-precision results                                        |



# LITERATURE REVIEW (19-20 / 20)

| SI | Title of Paper                                                                                                  | Publisher        | Methodology                                                                                                                                                                                                                                                               | Year | Advantages                                                                                                                                                                                                                                     | Disadvantages                                                                                                                                                                                                                                                                            |
|----|-----------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 | Neural radiance fields assisted by image features for UAV scene reconstruction                                  | Nature Portfolio | <b>(HIF-NeRF)</b> – UAV scene reconstruction using a hybrid CNN–Transformer image encoder, NeRF with an uncertainty-prediction branch to suppress dynamic regions, refined training losses, and Random Fourier Feature Encoding for better high-frequency detail capture. | 2025 | Delivers accurate UAV scene reconstruction, outperforms NeRF/NeRF-W/HA-NeRF on PSNR–SSIM–LPIPS, produces consistent geometry with fine details, generalizes well to novel views, and yields sharp depth edges even at long range.              | Struggles with sparse or repetitive textures, training cost rises sharply with more images or higher resolution, novel-view quality drops with low view overlap, and circular capture trajectories limit handling of lateral viewpoints                                                  |
| 20 | Towards the application of NeRFs and Gaussian Splitting for surface and textural representation in construction | CONVR-2025       | Comparative study of NeRF (Nerfacto) and Gaussian Splatting (Splatfacto) for construction material surface analysis, using 360° video capture, COLMAP-based SfM point clouds, and evaluation against ground truth for resolution, rendering quality, and defect detail.   | 2025 | <b>NeRF</b> excels at high-fidelity texture and surface detail for close, well-lit inspections, while <b>Gaussian Splatting</b> is faster (43 min vs 120 min), produces smoother, less noisy surfaces, and suits rapid, large-area assessments | <b>NeRF</b> struggles with distant/low-light textures and is compute-heavy, limiting large or dynamic use; <b>Gaussian Splatting</b> can blur sharp edges, miss fine details or introduce artifacts, and is sensitive to camera distance and angle, reducing precision defect detection. |

## 02: RESEARCH GAP



### Computational Latency

Traditional NeRF models suffer from high training and inference times, making real-time interactive rendering difficult for complex scenes.



### Sparse View Fidelity

Current algorithms degrade significantly in geometric accuracy when input 2D data is sparse or has limited angular coverage.



### Storage Overhead

3D Gaussian Splatting achieves high speeds but introduces massive disk storage requirements for high-resolution point cloud splats.

## 03: PROBLEM STATEMENT

The Core Challenge

How to achieve **photorealistic 3D scene generation** that supports real-time navigation while overcoming neural speed bottlenecks and visual occlusions?

Development of a robust pipeline to convert unstructured 2D photos into a high-fidelity, interactive 3D model with optimized geometric consistency.

# | 04: OBJECTIVE

## Hybrid Framework

Develop a unified framework integrating NeRF for volumetric geometry and Gaussian Splatting for rasterization.

## Real-time Rendering

Optimize inference to achieve rendering speeds exceeding 30 frames per second on standard hardware.

## Lighting Consistency

Address uneven lighting and shadow inconsistencies across multi-view input data using neural refinement.

## Data Compression

Implement efficient storage techniques to reduce Gaussian splat sizes without sacrificing visual fidelity.

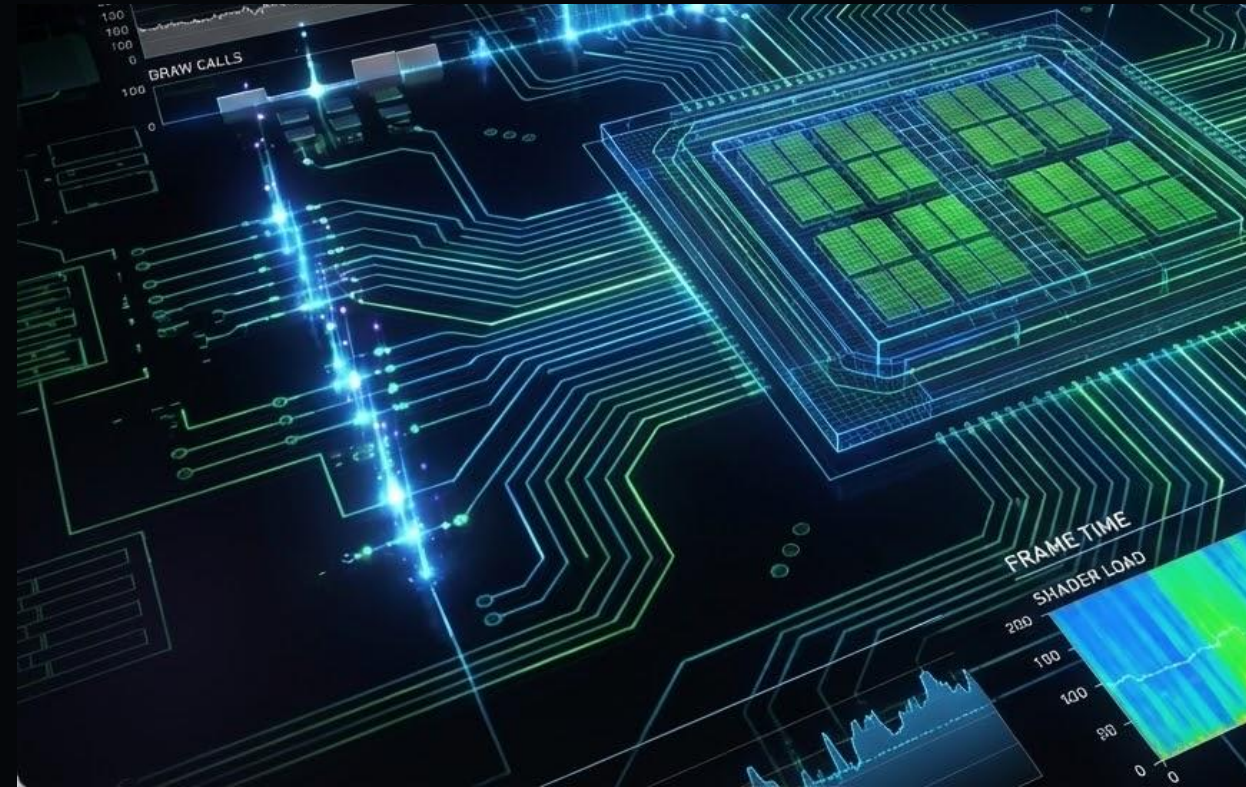
# NERF VS. SPLATTING

## Neural Radiance Fields

Implicit representation using neural networks. High photorealism but requires ray-marching per pixel, leading to slow rendering.

## 3D Gaussian Splatting

Explicit representation using 3D Gaussians. Allows for rasterization-based rendering, enabling high FPS interactive 3D viewing.



# CONCLUSION



Gaussian Splatting has revolutionized 3D scene representation by enabling real-time performance that NeRF could not reach alone.

Our study concludes that the future of 3D vision lies in hybrid pipelines that leverage neural learning for precision and explicit rendering for speed.

# THANK YOU

Questions & Discussion

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CEM Punnapra | Group Number - 05

Project: 3D Reconstruction from 2D Photos